

PCMC WardMitra (Sarathi) — AI Civic Grievance Orchestrator

System Architecture, Multi-Agent Dialogue Pipeline, Verification & Action Roadmap

Target Body: Pimpri Chinchwad Municipal Corp.	Version: 1.0.0 (Production Architecture)	Verified: September 2026
Test Suite: 50 / 50 Tests Passing (100%)	Pattern: Decoupled Multi-Agent Microservices	Runtime: FastAPI / Python 3.14.7

1. Executive Summary & Core Mission

PCMC WardMitra is an intelligent civic complaint orchestration platform engineered for the Pimpri Chinchwad Municipal Corporation. Unlike legacy portals or static chatbots, WardMitra is an active multi-agent AI system that natively handles conversational citizen dialogue in Marathi, Hindi, and English. It autonomously validates physical civic defects via Ultralytics YOLO11 computer vision, resolves exact GPS coordinates and administrative wards across all 32 PCMC divisions, detects fraudulent or duplicate submissions within a 50m spatial radius, and enforces automated 3-level municipal SLA escalation from Ward Officers to the Municipal Commissioner.

Architectural Highlights & Completed Core Capabilities:

- 9-Step Deterministic State Machine:** Citizen dialogue strictly follows an authenticated, guardrailed municipal workflow.
- Zero-Cost Free AI + Cloud Acceleration:** Smart hybrid router automatically switches between local Ollama (Llama 3.2, Rs. 0 cost) and OpenAI (GPT-4o-mini, <1s response) with zero-risk fallback.
- Precise Spatial Ward Engine:** Full PCMC bounding box validation with 32 administrative wards and landmark geocoding.
- Indic Voice Support:** Integrated Bhashini STT pipeline enabling citizens to speak complaints in vernacular Marathi/Hindi.

2. Complete 9-Step Citizen Conversation & Registration Flow

The dialogue flow is governed by a stateful orchestrator that prevents invalid submissions while ensuring natural citizen engagement:

#	Stage	Underlying AI Agent / Engine	Operational Function & Municipal Guardrails
1	Greeting & Language	ConversationalService Language Detection	Detects Marathi, Hindi, or English. Greets citizen warmly and sets civic context without registering premature tickets.
2	Intent & Category	NLPAgent + Heuristics	Categorizes issue (Pothole, Garbage, Streetlight, Water, Drainage, Encroachment, or Civic Knowledge Inquiry).
3	Interactive Clarification	Clarification Engine	If citizen provides ambiguous keywords (e.g. 'light'), bot presents interactive options (Streetlight vs Traffic Signal).
4	YOLO11 Evidence Check	ImageAgent (YOLO11 Vision)	Validates uploaded photo. Confirms actual civic defect. Rejects selfies, pets, documents, or deceptive images.
5	Text-Image Match	ImageAgent + Orchestrator	Ensures citizen's written complaint matches the visual evidence (e.g. pothole text requires pothole photo evidence).
6	Geo & Duplicate Check	GeoAgent + FraudAgent	Extracts GPS/landmark. Maps to 1 of 32 PCMC wards. Verifies 50m Haversine radius for duplicate tickets to prevent spam.
7	Severity & Priority	PriorityAgent + SLA Engine	Detects emergencies (open manholes, live wires, gas leaks) and upgrades SLA from normal (48h) to critical emergency (4h).
8	Interactive Preview	ConversationalService	Renders ticket preview card with Category, Ward Name, Officer in charge, SLA Deadline, and GPS before confirmation.
9	Ticket Registration	GrievanceOrchestrator	Issues unique collision-free Ticket ID (PCMC-YYYYMMDD-XXXX). Dispatches notification and starts SLA escalation countdown.

3. Technical Architecture & Micro-Agent Implementation

WardMitra is structured into modular, independently scalable services that isolate business rules, ML inference, and persistence:

Module / Component	Code Location	Architectural Capabilities & Implementation Details
Grievance Orchestrator	app/orchestrator/orchestrator.py	Central state-machine coordinator. Executes sequential validation pipelines, verifies ticket ID uniqueness against DB collision, and manages persistence.
Smart Hybrid LLM Client	app/clients/llm_client.py	Tier 1: Instant regex/rules. Tier 2: Local Ollama (Llama 3.2, Rs. 0 cost). Tier 3: Cloud OpenAI (GPT-4o-mini, <1s latency). Graceful zero-crash fallback.
Vision Agent (YOLO11)	app/agents/image_agent.py	Ultralytics YOLO11 deep learning for civic defect classification. Evaluates confidence thresholds, detects spoofing, and matches text complaints.
GIS & 32-Ward Agent	app/agents/geo_agent.py	Enforces PCMC boundary (18.57-18.72° N, 73.74-73.92° E). Contains all 32 PCMC wards with assigned officers. Resolves text landmarks (Wakad, Bhosari, Nigdi).
Duplicate & Fraud Agent	app/agents/fraud_agent.py	Haversine 50m spatial clustering. Flags repeat submissions, computes user fraud score, and increments duplicate count instead of spamming departments.
3-Level SLA Escalation	app/services/escalation_service.py	Automated background cron via APScheduler. Level 1: Ward Officer (24h) → Level 2: Executive Engineer (48h) → Level 3: Municipal Commissioner (72h).
Bhashini Indic STT & TTS	app/services/bhashini_stt_service.py app/services/bhashini_tts_service.py	Bhashini ASR (voice input with WhatsApp OGG/Opus auto-conversion to 16kHz WAV) + Bhashini TTS (voice output in Marathi, Hindi, English).
Civic Knowledge Base	app/services/human_persona_service.py	Answers non-complaint civic queries: birth/death certificate process, property tax payment portals, garbage van schedules, and emergency helplines.
Content Moderation	app/services/moderation_service.py	Multilingual profanity filter (Marathi/Hindi/English) and optional AWS Rekognition visual safety filter to protect municipal records.

4. Quality Assurance, Test Coverage & Performance Metrics

The platform includes an automated pytest suite containing 50 end-to-end integration and unit tests, achieving 100% pass status:

Test Suite: pytest tests/ -v	Total Scenarios: 50 Test Cases	Passing Rate: 50 / 50 (100% PASSED)	Failed / Broken: 0 (Zero Errors)
Execution Time: 50.68 seconds	Python: 3.14.7 FastAPI 0.141	SLA Monitor: Verified Clean	Database: SQLite ACID Verified

Category Tested	Test File	Key Behaviors Validated
15 WardMitra Scenarios	tests/test_wardmitra_scenarios.py	Clarifications (light), categories (drainage), Marathi/English inputs, invalid images, pothole YOLO verification, duplicate 50m detection, emergency speedup.
Human Persona & KB	tests/test_human_persona_and_kb.py	Birth certificate queries, property tax URLs, garbage schedule, helpline numbers, warm Marathi tone, dynamic ward detection (Wakad, Bhosari, Nigdi).
Voice STT (Bhashini)	tests/test_bhashini_stt.py	Valid audio transcription, base64 payload handling, empty audio validation, multimodal voice complaint registration.
Moderation & Safety	tests/test_moderation.py	Multilingual profanity interception, image safety checks, complaint rejection on unsafe content, clean civic submissions.
End-to-End SLA & Escalation	tests/test_end_to_end.py	Full complaint lifecycle, 3-level escalation progression (Level 1→2→3), spam blocking, repeat count increment, APScheduler cycle.

5. Operational Action Plan: What Should Be Done (Roadmap)

To take the validated WardMitra system to city-wide citizen production, the following execution roadmap is established across 4 practical phases:

Phase & Priority	Focus Area	Action Items & Deliverables
Phase 1 IMMEDIATE (Days 1–3)	API Credentials & Hybrid LLM Tuning	• Configure OpenAI Key (Optional): Insert OPENAI_API_KEY in .env to activate <1s human-like Marathi conversation. Ollama remains 100% active as zero-risk fallback. • Configure Bhashini Credentials: Add Bhashini User ID and Pipeline keys in .env for live voice transcription. • Sanity Check: Verify live interactive Swagger docs on http://localhost:8000/docs.
Phase 2 HIGH (Week 1–2)	Omnichannel Citizen Frontend	• Citizen Web Chat Widget: Deploy responsive citizen portal UI with interactive suggestion chips (e.g. 'Streetlight' vs 'Traffic Light'), audio mic recording, camera upload, and Leaflet GPS map picker. • WhatsApp Grievance Bot: Connect Meta Cloud API / Gupshup webhook directly to /api/v1/chat/complaint to allow citizens to file complaints in Marathi over WhatsApp. • Mobile App Integration: Connect Flutter / React Native citizen app to the existing FastAPI endpoints.
Phase 3 MEDIUM (Week 2–3)	Ward Officer Portal & Resolution Verification	• Officer Kanban Dashboard: Real-time ticket management board grouped by Ward (1 to 32) and Priority (Emergency, High, Normal). • Visual Resolution Verification: Field officers must upload an 'After' photograph upon fixing the defect. YOLO11 verifies defect clearance before ticket closure. • Citizen SMS Alerts: Notify citizens when status updates from Submitted → In Progress → Resolved.
Phase 4 ENTERPRISE (Week 3–4)	Production Hardening & Municipal Cloud	• PostgreSQL + PostGIS Migration: Upgrade SQLite to enterprise PostgreSQL with PostGIS for spatial indexing. • Docker & Kubernetes: Containerize FastAPI, APScheduler, and YOLO11 via Docker Compose for PCMC cloud. • Security & Gateway: Enforce HTTPS TLS 1.3, rate-limiting per citizen phone, and CORS policies.

6. Key API Endpoints Reference

All services are exposed as standardized REST endpoints ready for frontend, mobile, or WhatsApp integration:

Endpoint	Method	Function	Payload & Parameters
/api/v1/chat/complaint	POST	Unified Citizen Chatbot	message (str), session_id (str), optional image, optional audio
/api/v1/complaints/submit	POST	Direct Form Complaint	category, description, latitude, longitude, citizen_phone, image
/api/v1/voice/stt	POST	Bhashini Voice STT	audio_file or audio_base64 (WAV/MP3/M4A/OGG), language (mr/hi/en)
/api/v1/voice/tts	POST	Bhashini Voice TTS	text (str), language (mr/hi/en), gender (female/male). Returns base64 WAV.
/api/v1/complaints/status/{id}	GET	Live Ticket Tracking	Path param: id (e.g. PCMC-20260919-AB12). Returns status, SLA deadline.
/api/v1/escalation/run-check	POST	Manual SLA Scan	Scans overdue complaints and triggers 3-level escalation notifications immediately.
/health	GET	System Health Check	Validates Database connection, YOLO11 Vision Model status, and LLM Provider.

Preparedness & System Readiness Summary:

The PCMC WardMitra AI backend is 100% operational, architecturally sound, and fully verified across all 50 test scenarios. It eliminates civic grievance spam, eliminates manual department dispatching, prevents duplicate processing via 50m spatial clustering, and enforces municipal accountability through automated 3-level escalation. The system is ready for frontend deployment and live citizen onboarding.